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HRP-AI: A Web Application for Hypertension Risk Prediction with Calibrated Explainable AI

Jonathan Ralph F. Baes

Abstract—Hypertension remains a major public health con cern in the Philippines and requires scalable screening tools that are both interpretable and probability-aware. This paper presents HRP-AI, a web-based decision-support application for predicting clinical hypertension risk among Filipino adults using calibrated explainable machine learning. Using the 2015 DOST FNRI National Nutrition Survey, a dataset of 151,189 adults was constructed, including 22,988 hypertension-positive cases based on the ≥ 140/90 mmHg clinical threshold. Eight machine learning algorithms were evaluated under multiple imbalance handling configurations using a two-stage randomized optimiza tion strategy. XGBoost with class weights was selected as the final predictive engine because it provided the strongest clinical compromise between sensitivity, discrimination, and calibration compatibility. At the standard 0.50 decision threshold, the model achieved 76.87% accuracy, 76.51% recall, an F1-score of 0.5014, and an AUC of 0.8469. Under the selected 0.35 screening threshold, recall increased to 87.88%. Venn-Abers calibration reduced the Expected Calibration Error to 0.0035. The final pro totype integrates calibrated risk estimates, uncertainty bounds, SHAP and LIME explanations, and Wachter-style counterfactual suggestions. The results show that non-linear machine learning models can be adapted into transparent, probability-aware, and exploratory hype screening tools when discrimination, calibration, explainability,

practical issues limit the adoption of machine learning systems in clinical deci sion support. First, high-performing models such as gradient boosting ensembles are often considered black boxes, making

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their predictions difficult to interpret. Second, discriminative performance does not guarantee probability reliability; a model can rank patients well while producing poorly calibrated risk estimates. This is problematic in healthcare because risk scores are often interpreted as probabilities.

Explainable artificial intelligence (XAI) methods such as SHAP and LIME improve transparency by identifying the features that influence a prediction. Probability calibration methods such as Platt scaling, isotonic regression, and Venn Abers predictors improve the reliability of predicted probabil ities. This study integrates these ideas into HRP-AI, a web application for hypertension risk prediction with calibrated explainable AI. The system is designed not as a replacement for clinical diagnosis but as an exploratory decision-support type for risk screening.

The main contributions of this work are as follows: • a

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